S8703A Functional KPI Toolset

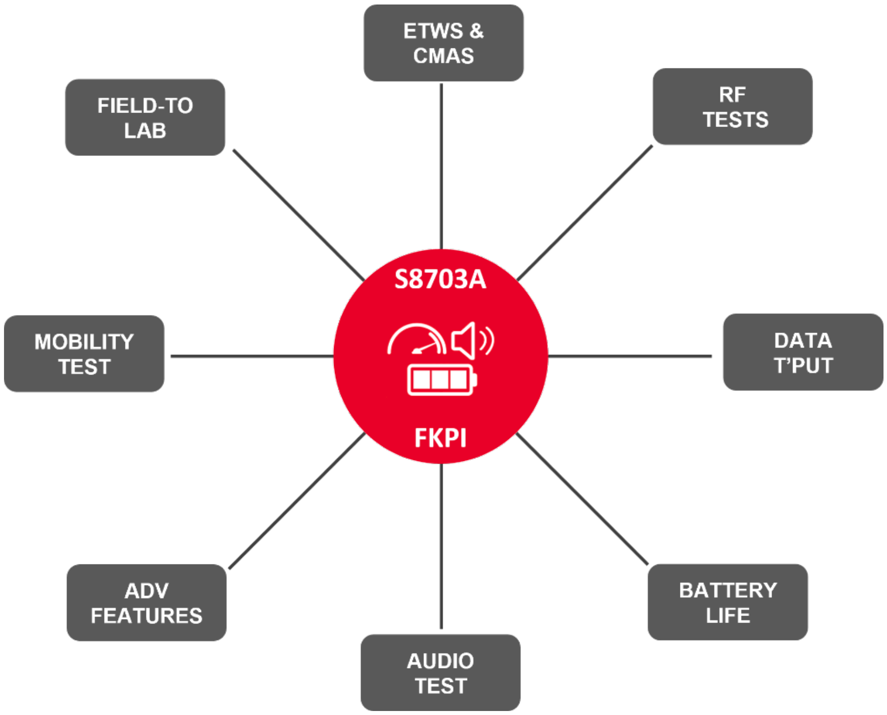
Functional and Performance testing - made easy

The success of 5G networks depends upon their ability to provide optimal user experience. Network operators, Chipset manufacturers, Device manufacturers, test houses and system integrators all have to role to play to achieve this, and the Keysight S8703A Functional KPI (FKPI) Toolset provides engineers within these organizations, a cost effective, scalable, and easy to use solution to test and verify functionality and performance of 5G devices. Early detection of functional test issues is paramount for cost savings and for successful rollout of high-quality devices.

KEYSIGHT

Introduction

Leveraging the power, flexibility, and test coverage of the market leading UXM 5G Wireless Test Platform, the Keysight S8703A FKPI provides an automated benchtop solution for the verification of 5G, 4G, WLAN and NB-NTN user experience. Functional test areas addressed are data throughput, battery life, audio / video functionality, mobility, emergency services, location-based services and WLAN performance making it a must have in labs focusing on design verification, performance optimization and device repair.



The FKPI also provides advanced capabilities such as Field-to-Lab, the ability to recreate in labs, the faults that have been observed in the network. UE logs are utilized for this purpose and the FKPI then provides tools to fix the problems and then perform thorough tests to ensure resolution. Besides, FKPI provides Eggplant based UE automation, Cable Replacement Method, tests for dual SIM deployments and many other features.

Based on the S8711A Test Application and S8714A RF Application Framework

Keysight's 5G Test Application (TA) Framework & RF Application (RA) allow both, touch-based and remote control of the UXM 5G network emulator. The various operation modes support a wide range of tests, from non-signaling for UE Calibration to full signaling tests, providing a comprehensive set of features for engineers designing RF components and devices. The framework supports 4G LTE, NB-NTN as well as both of the 5G deployment modes, Non-Standalone (NSA) and Standalone (SA). The full signaling test mode enables users to perform RF measurements while a call is in progress. Besides the cellular standards, the RA Framework also supports WLAN performance testing for standards ranging from 802.11b/g through to 802.11be.

The FKPI Toolset uses these underlying frameworks to control the UXM 5G network emulator and provide a benchtop suite of fully automated functional tests, where users can easily create campaigns and run them under accurate and repeatable user scenarios.

Figure 1. RF Measurement results on Keysight's 5G RF Application Framework

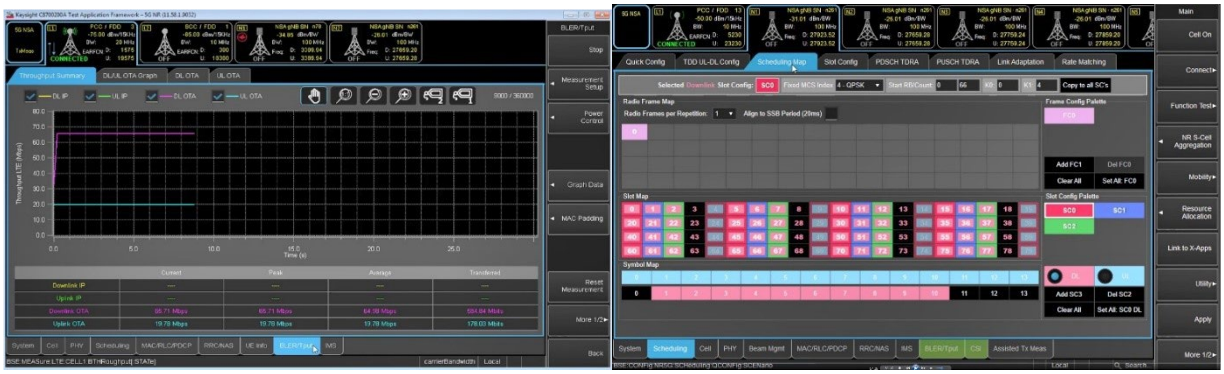


Figure 2. Throughput measurement results on Keysight's 5G Test Application Framework

Functional KPI Toolset – Key features

Keysight's Functional KPI Toolset extends the capabilities of the 5G Test Application and 5G RF Application by providing:

- An intuitive and easy-to-use graphical user interface for creating, configuring, and running test campaigns.
- A suite of fully automated functional test for suite covering the following:
- **Data throughput testing**
  - 5G NR Standalone / Non-standalone modes
  - Multiple data transfer protocols
  - FR1 & FR2 support
  - Multi-carrier Support
  - 4G LTE support
- **Battery life testing**
  - Current drain analysis for voice and data use cases
  - Battery life estimation
- **Audio functional test**
  - VoLTE / VoNR
  - Audio functional tests
  - IMS/SIP IR-92 Compliance tests
  - Mobile Originated / Mobile Terminated calls
  - Performance testing under different network and radio conditions
- **Dual-UE testing**
  - Execute functional tests simultaneously on two devices using one UXM 5G network emulator.
- **Dual-SIM testing**
  - Dual SIM / Dual Standby (DSDS) and Dual SIM / Dual Active (DSDA) Scenarios.
  - Requires only one UXM 5G network emulator now

- **Mobility testing**
  - Verification of Handover with data continuity
  - Cell selection / Reselection / Re-direction
  - Support for channel emulation
- **Field-to-lab testing**
  - Import field log configurations and parameters for functional test on UXM 5G.
  - Use automated test script creation to save time and effort
- **LBS A-GNSS testing**
- **Emergency Services**
  - Signaling & messaging for Commercial Mobile Alert System (CMAS), Earthquake and Tsunami Warning System (ETWS)
- Additional test case packages for advanced feature testing and operator regulatory compliance are available as well: -
  - **Bandwidth Parts (BWP) tests**
  - **Dynamic Spectrum Sharing (DSS) tests**
  - **Physical layer tests**
  - **WLAN performance tests**
  - **NB-NTN tests**
  - **Regulatory test case packages**

- Optimized test execution times, enabling quick inspection of the Functional Test of devices.
- A report generator in different formats to summarize the results of test campaigns.
- Support for 4G LTE, NB-NTN, 5G (NSA & SA modes) and WLAN in the same network emulator, providing a small footprint benchtop solution.
- State-of-the art logging, visualization and debugging tools.
- Flexible licensing options and tools

Campaign creation, configuration and management

The versatility of the FKPI Toolset interface allows users with little to no programming experience as well as more experienced users to easily create, setup and execute test campaigns to verify their devices. There are two modes of operation, Runtime mode and Development mode.

In the Runtime mode, users have access to ready sample testcases with key parameters such as bandwidth, channel, MCS exposed to the user, for quick test plan setup and execution in just a few clicks.

The Development mode is for more experienced users who have the need to create more customized test plans. This mode provides access to configure lower-level test steps with complete flexibility to setup and build custom test cases, while also offering an easy to use and intuitive graphical user interface.

Each test in the campaign may be executed over a range of test conditions specified by the user, resulting in a test generated for each individual test condition. Test conditions provide access to a range of test-specific parameters, including bandwidth, modulation scheme, power levels, and the limits measurement values against which the pass/fail verdict is assessed. This provides users with the flexibility to focus campaigns on specific conditions that need further debugging, to quickly complete the verification of a device's functionality.

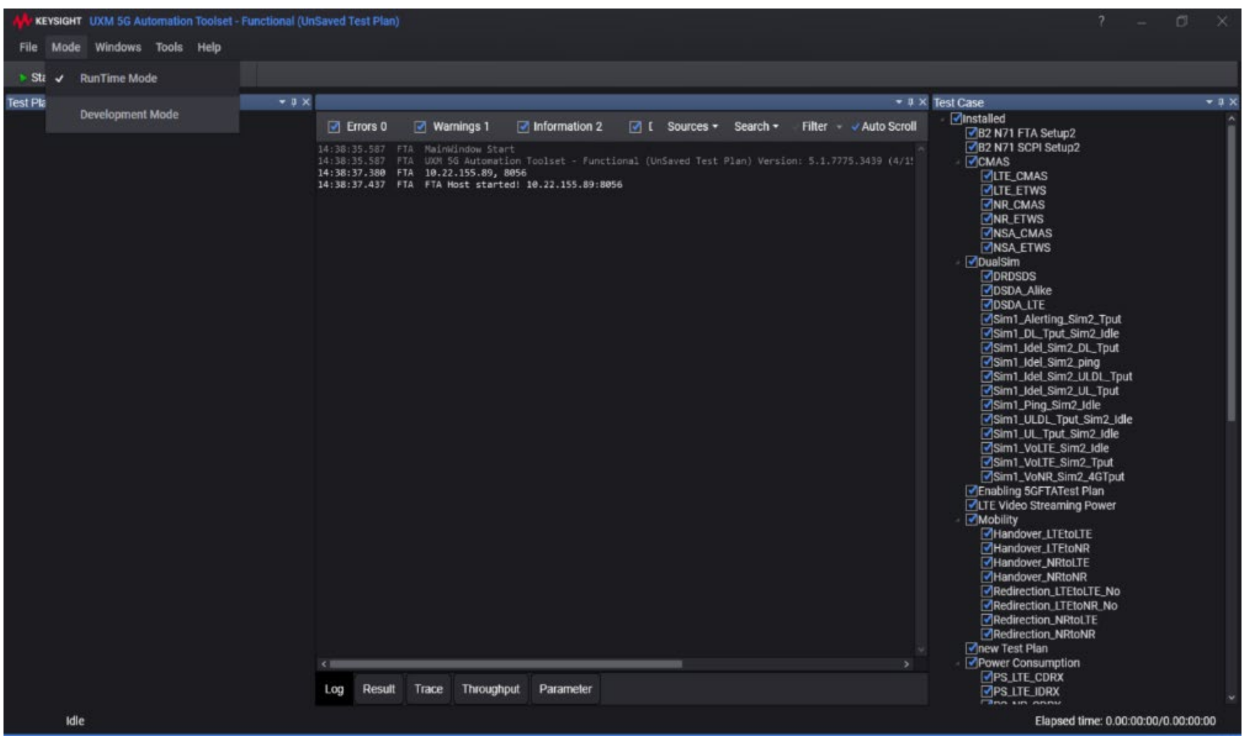


Figure 3. Campaign created by selecting available test steps in Runtime Mode

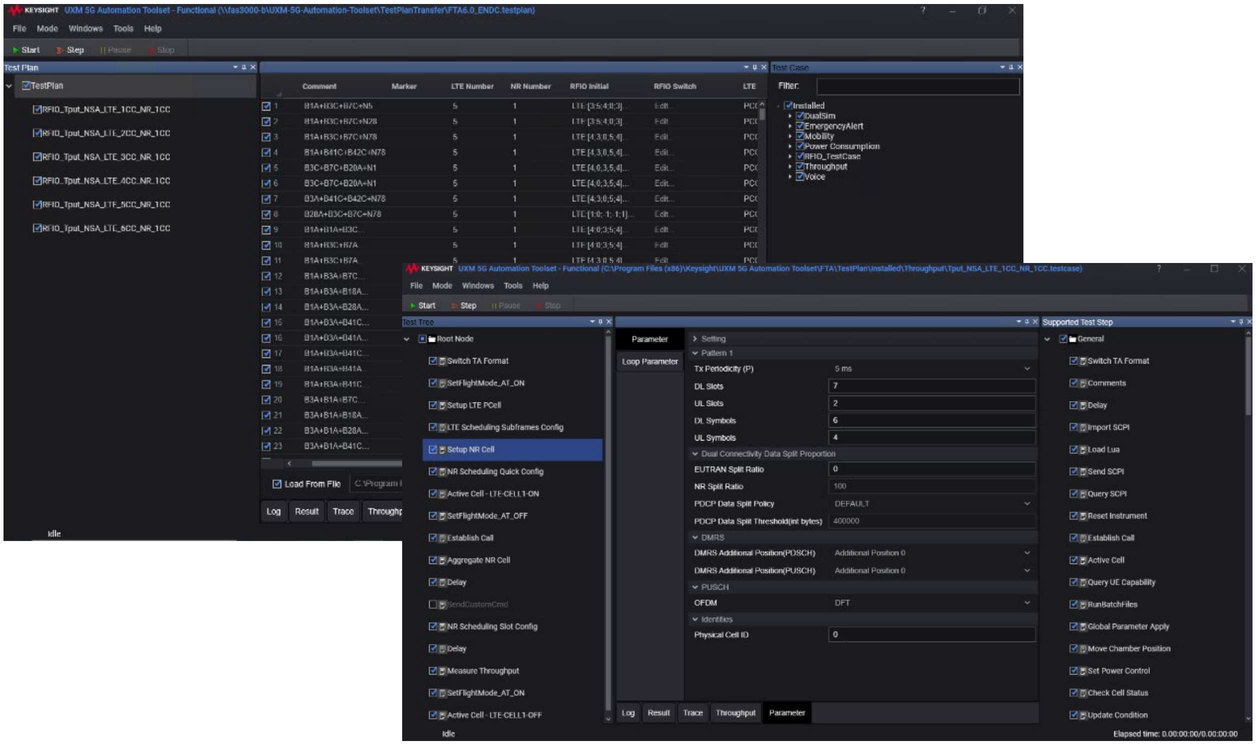


Figure 4. Campaign created by selecting available test steps in Operator Mode

Result analysis

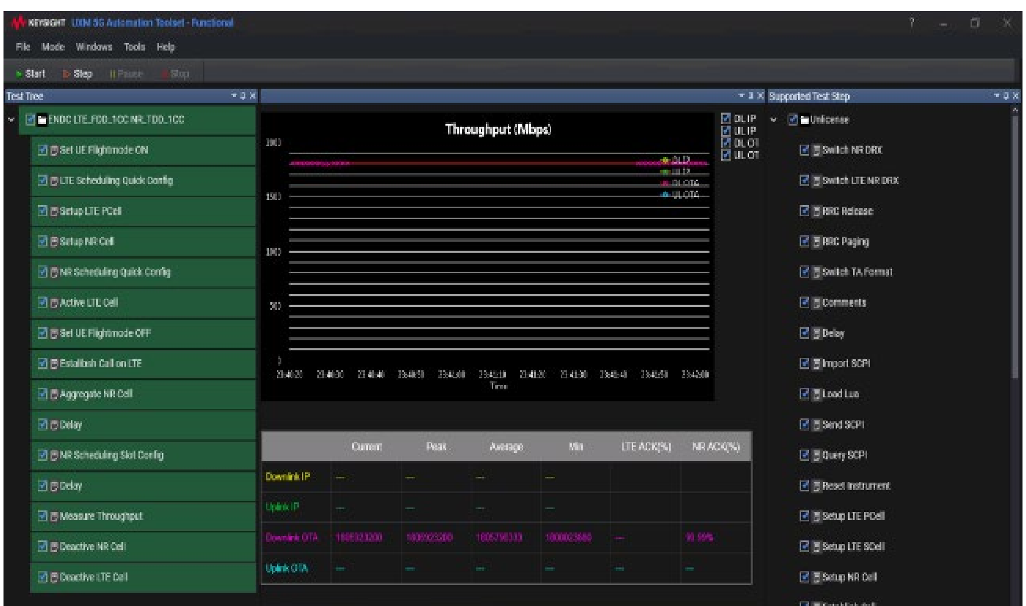
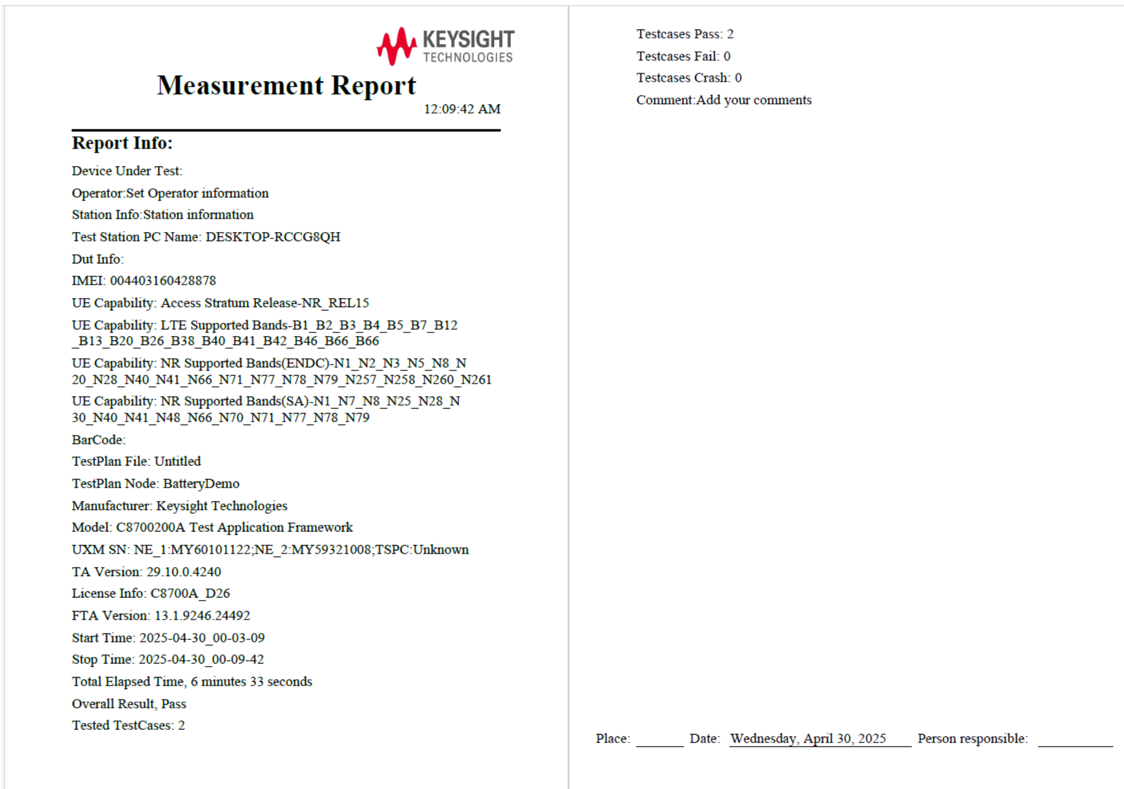


Figure 6. Successful test steps with color coded verdict

The Result List area along with the bold color verdict indicators on test steps, provides real-time information about the test campaign execution progress with relevant information about each individual test result, including a pass/fail verdict for each test based on the user-defined verdict conditions. Users can easily also switch between tabs and view any available graphical measurement results

Report generation

For post-campaign analysis, the Functional KPI Toolset allows the generation of summary reports in csv, xlsx and PDF formats



KEYSIGHT TECHNOLOGIES

Measurement Report

12:09:42 AM

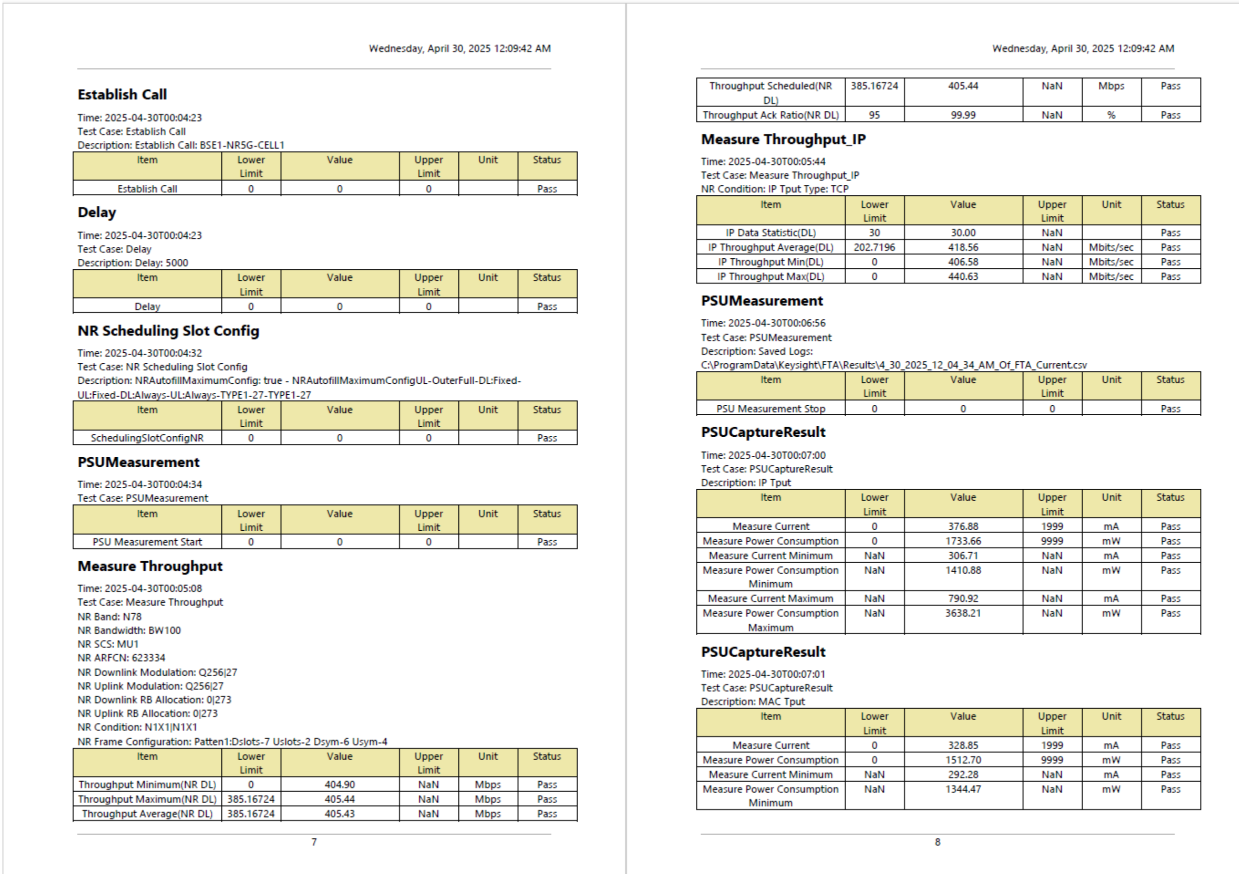
Report Info:

Device Under Test:
Operator:Set Operator information
Station Info:Station information
Test Station PC Name: DESKTOP-RCCG8QH
Dut Info:
IMEI: 004403160428878
UE Capability: Access Stratum Release-NR_REL15
UE Capability: LTE Supported Bands-B1_B2_B3_B4_B5_B7_B12_B13_B20_B26_B38_B40_B41_B42_B46_B66_B66
UE Capability: NR Supported Bands(ENDC)-N1_N2_N3_N5_N8_N20_N28_N40_N41_N66_N71_N77_N78_N79_N257_N258_N260_N261
UE Capability: NR Supported Bands(SA)-N1_N7_N8_N25_N28_N30_N40_N41_N48_N66_N70_N71_N77_N78_N79
BarCode:
TestPlan File: Untitled
TestPlan Node: BatteryDemo
Manufacturer: Keysight Technologies
Model: C8700200A Test Application Framework
UXM SN: NE_1:MY60101122;NE_2:MY59321008;TSPC:Unknown
TA Version: 29.10.0.4240
License Info: C8700A_D26
FTA Version: 13.1.9246.24492
Start Time: 2025-04-30_00-03-09
Stop Time: 2025-04-30_00-09-42
Total Elapsed Time, 6 minutes 33 seconds
Overall Result, Pass
Tested TestCases: 2

Testcases Pass: 2
Testcases Fail: 0
Testcases Crash: 0
Comment:Add your comments

Place: ______ Date: Wednesday, April 30, 2025 Person responsible: __________

Figure 7. Example of overall measurement report in pdf format



Wednesday, April 30, 2025 12:09:42 AM

Establish Call

Time: 2025-04-30T00:04:23
Test Case: Establish Call
Description: Establish Call: BSE1-NR5G-CELL1

Item	Lower Limit	Value	Upper Limit	Unit	Status
Establish Call	0	0	0		Pass

Delay

Time: 2025-04-30T00:04:23
Test Case: Delay
Description: Delay: 5000

Item	Lower Limit	Value	Upper Limit	Unit	Status
Delay	0	0	0		Pass

NR Scheduling Slot Config

Time: 2025-04-30T00:04:32
Test Case: NR Scheduling Slot Config
Description: NRAutofillMaximumConfig: true - NRAutofillMaximumConfigUL-OuterFull-DL:Fixed-UL:Fixed-DL:Always-UL:Always-TYPE1-27-TYPE1-27

Item	Lower Limit	Value	Upper Limit	Unit	Status
SchedulingSlotConfigNR	0	0	0		Pass

PSUMeasurement

Time: 2025-04-30T00:04:34
Test Case: PSUMeasurement

Item	Lower Limit	Value	Upper Limit	Unit	Status
PSU Measurement Start	0	0	0		Pass

Measure Throughput

Time: 2025-04-30T00:05:08
Test Case: Measure Throughput
NR Band: N78
NR Bandwidth: BW100
NR SCS: MU1
NR ARFCN: 623334
NR Downlink Modulation: Q256|27
NR Uplink Modulation: Q256|27
NR Downlink RB Allocation: 0|273
NR Uplink RB Allocation: 0|273
NR Condition: N1X1|N1X1
NR Frame Configuration: Patten1:Dslots-7 Uslots-2 Dsym-6 Usym-4

Item	Lower Limit	Value	Upper Limit	Unit	Status
Throughput Minimum(NR DL)	0	404.90	NaN	Mbps	Pass
Throughput Maximum(NR DL)	385.16724	405.44	NaN	Mbps	Pass
Throughput Average(NR DL)	385.16724	405.43	NaN	Mbps	Pass

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Wednesday, April 30, 2025 12:09:42 AM

Throughput Scheduled(NR DL)	385.16724	405.44	NaN	Mbps	Pass
Throughput Ack Ratio(NR DL)	95	99.99	NaN	%	Pass

Measure Throughput_IP

Time: 2025-04-30T00:05:44
Test Case: Measure Throughput_IP
NR Condition: IP Tput Type: TCP

Item	Lower Limit	Value	Upper Limit	Unit	Status
IP Data Statistic(DL)	30	30.00	NaN		Pass
IP Throughput Average(DL)	202.7196	418.56	NaN	Mbits/sec	Pass
IP Throughput Min(DL)	0	406.58	NaN	Mbits/sec	Pass
IP Throughput Max(DL)	0	440.63	NaN	Mbits/sec	Pass

PSUMeasurement

Time: 2025-04-30T00:06:56
Test Case: PSUMeasurement
Description: Saved Logs:
C:\ProgramData\Keysight\FTA\Results\4_30_2025_12_04_34_AM_Of_FTA_Current.csv

Item	Lower Limit	Value	Upper Limit	Unit	Status
PSU Measurement Stop	0	0	0		Pass

PSUCaptureResult

Time: 2025-04-30T00:07:00
Test Case: PSUCaptureResult
Description: IP Tput

Item	Lower Limit	Value	Upper Limit	Unit	Status
Measure Current	0	376.88	1999	mA	Pass
Measure Power Consumption	0	1733.66	9999	mW	Pass
Measure Current Minimum	NaN	306.71	NaN	mA	Pass
Measure Power Consumption Minimum	NaN	1410.88	NaN	mW	Pass
Measure Current Maximum	NaN	790.92	NaN	mA	Pass
Measure Power Consumption Maximum	NaN	3638.21	NaN	mW	Pass

PSUCaptureResult

Time: 2025-04-30T00:07:01
Test Case: PSUCaptureResult
Description: MAC Tput

Item	Lower Limit	Value	Upper Limit	Unit	Status
Measure Current	0	328.85	1999	mA	Pass
Measure Power Consumption	0	1512.70	9999	mW	Pass
Measure Current Minimum	NaN	292.28	NaN	mA	Pass
Measure Power Consumption Minimum	NaN	1344.47	NaN	mW	Pass

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Figure 8. Example of details in the measurement report

Enabling automation

Shown below, is a typical test setup for device test. FKPI provides a stable API for automated control of all the components, as well as allowing other software/automation suite to call on FKPI.

Manual execution of tests incurs a lot of time and resource overhead. Enabling automation is a key delivering faster and more accurate results. Automation enhances the effectiveness, expands the test coverage, and enhances the speed and efficiency of the test process.

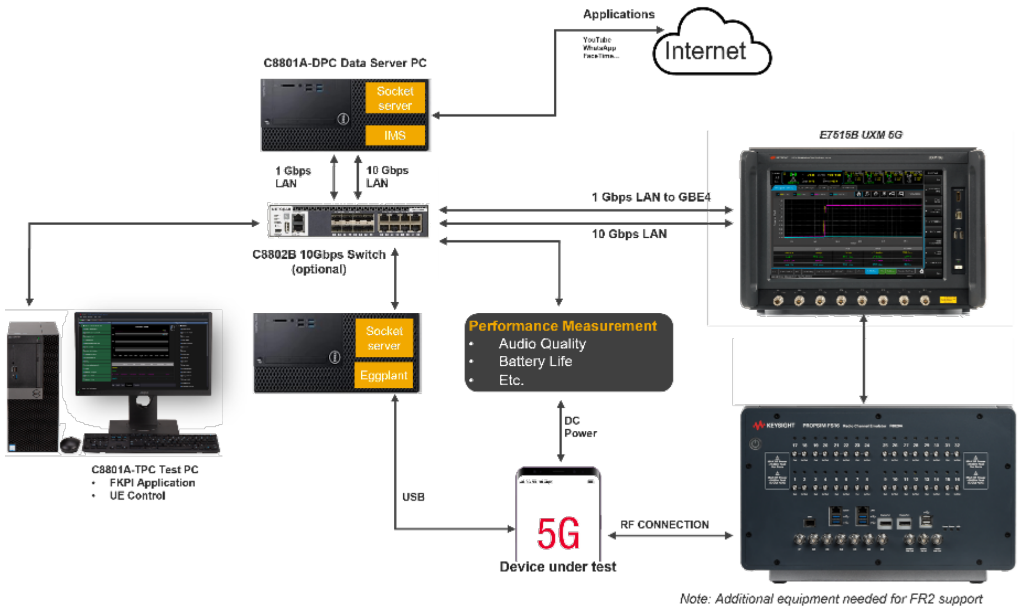

Figure 9. Set-up for test automation

- Application features are aimed towards setting up test cases/campaigns and are speed optimized to achieve fast execution and report generation.
- Integration with **Eggplant** offers a unique UE automation capability to control the DUT during the test process. Several UE functions like Airplane Mode, Voice call testing, Application performance can be automated by using Eggplant scripts and commands provided as test steps in Functional KPI Toolset. This is supported for both iOS and Android based devices.

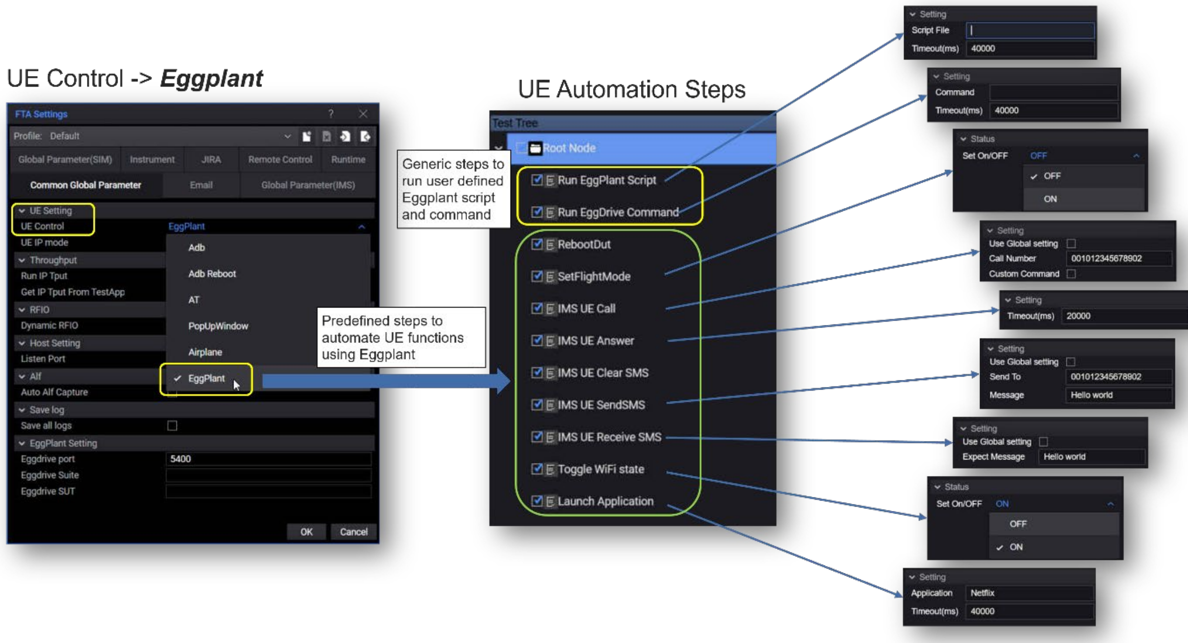

Figure 10. Eggplant automation

Automated band combination testing

Automated band combination test mode enables the users to test the DUT across different band combinations in an easy and effective way. When testing around 1000 different band combinations for the DUT, this feature/solution is hugely effective at minimizing time and manual effort required.

Functional KPI Toolset enables the testing of different band combinations by allowing the users to import a .CSV file that dictates the bands to loop the test on. This file contains information about band combinations, BW, channel etc.

Users can automate the testing of RF performance and different functionalities like Throughput, Mobility, VoLTE/NR across different band combos.

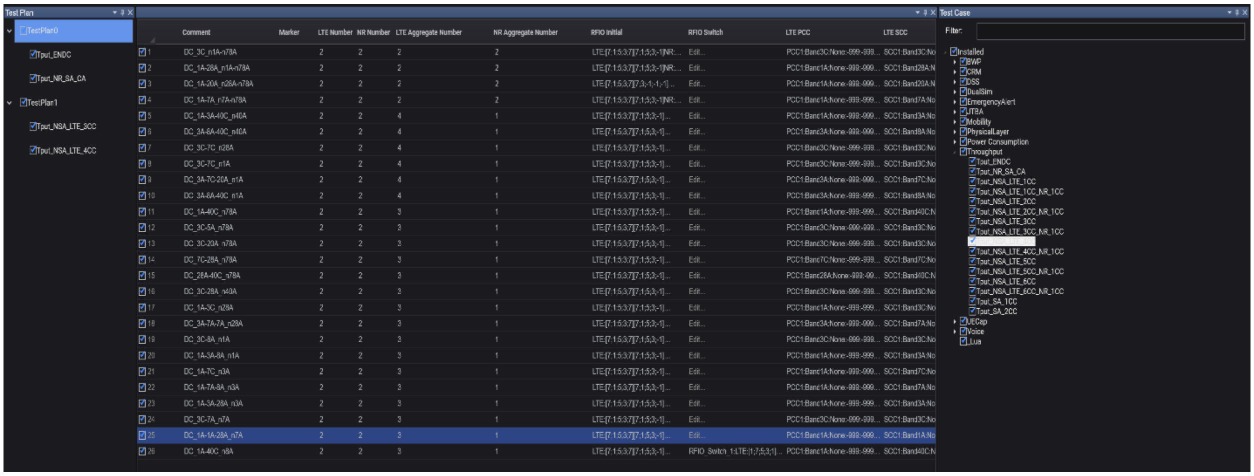

Figure 11. Automated Band combination testing using Functional KPI Toolset

Dynamic RFIO mapping

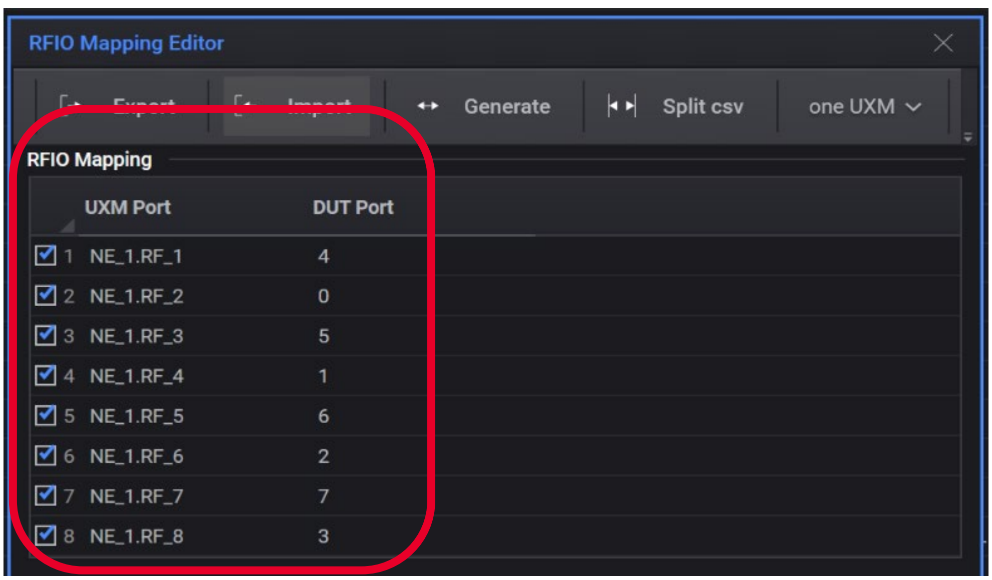

Figure 12. RFIO Mapping Editor feature

Functional KPI Toolset can also switch UXM RFIO ports dynamically according to the band combination being tested, which makes this a truly automated test without the need of re-cabling and minimizing errors and delays dung the test. Users can also choose to use the default RFIO mapping or provide their own RFIO mapping file using the import feature.

Data throughput testing

One of the key focus areas for 5G NR is Enhanced Mobile Broadband (eMBB) which promises speeds up

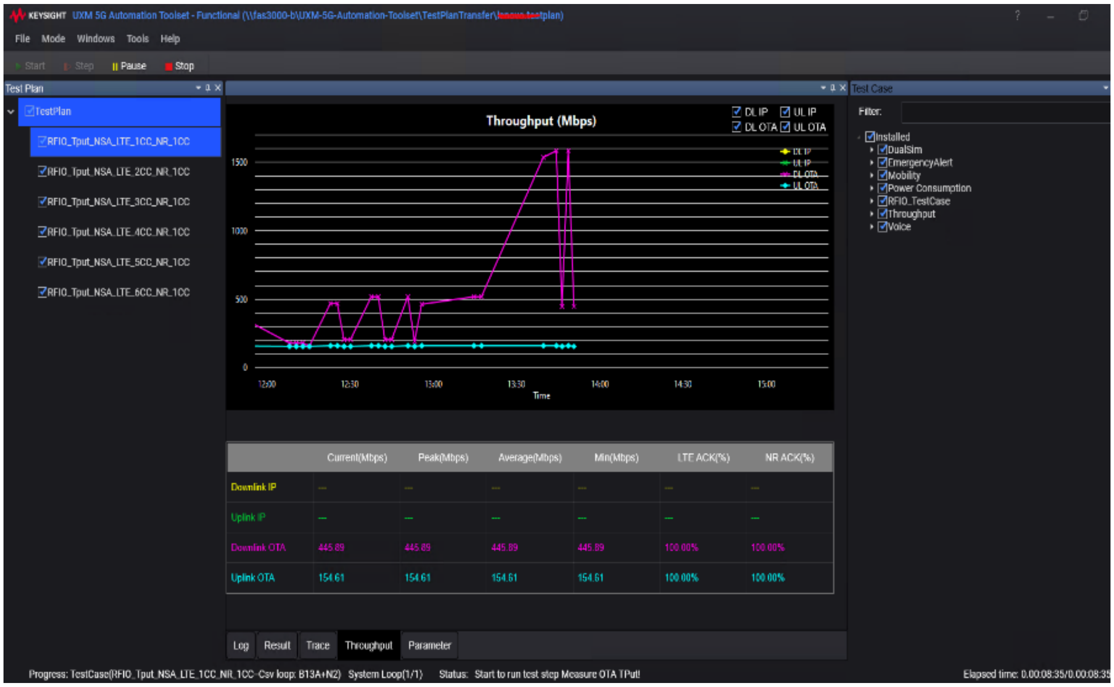

Figure 13. Data Throughput testing with Functional KPI

to 10 Gbps peak throughput, 1 Gbps throughput in high mobility scenarios and at a lower cost per bit per hertz transmitted. The technology also requires the need to support wider bandwidths of up to 100 MHz with multiple bands, carriers, and MIMO streams, creating a very challenging set of features and environment for devices.

The S8703A Functional KPI Toolset along with E7515B/W/P UXM 5G Wireless Test Platform gives engineers a powerful and flexible tool to simplify the testing and verification of their device performance for data throughput, supporting both IP and MAC layer throughput. Users have the ability with just a few clicks to easily setup accurate and repeatable real-user scenarios while leveraging sample test cases to get your test up and running in no time.

Smart Scheduling for data throughput testing

Smart Scheduling is a unique feature designed to help users simplify the scheduling configuration for different duplex mode and frame structures. 5G has very flexible PHY configuration thus making the scheduling very complicated due to variables in the PHY layer. With the S8703A FKPI Smart Scheduling function, users only need to specify the duplex mode (FDD or TDD), frame structures (including one cycle, dual cycle and different special frame structure), and FKPI tool takes care of all necessary parameters required to configure the scheduling settings in the Test Application accordingly. This feature not only simplifies the setting but also makes it possible to use a single test case for different band / frame combinations, providing a very easy means to maintain test cases.

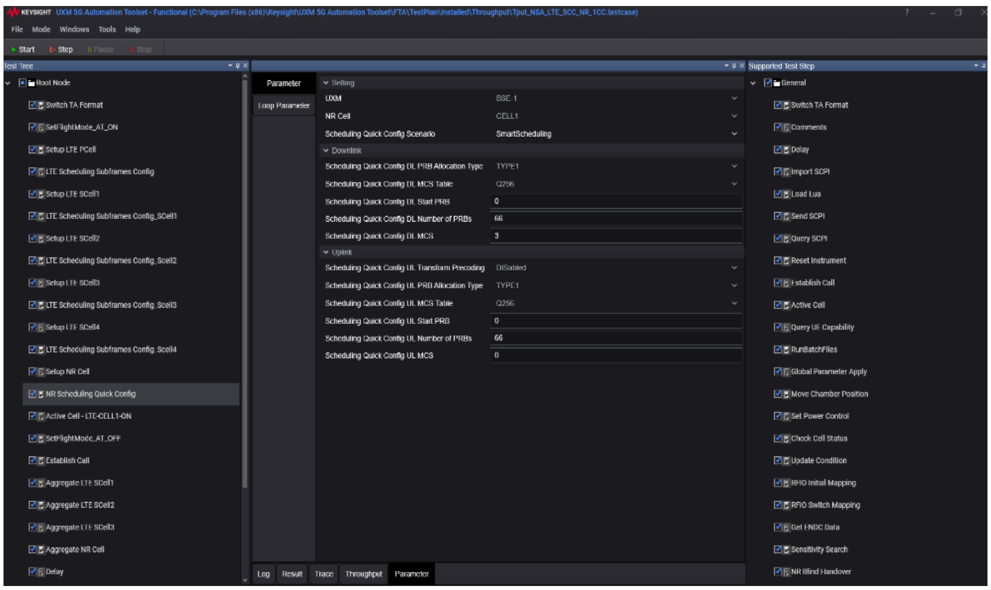

Figure 15. Smart Scheduling configuration

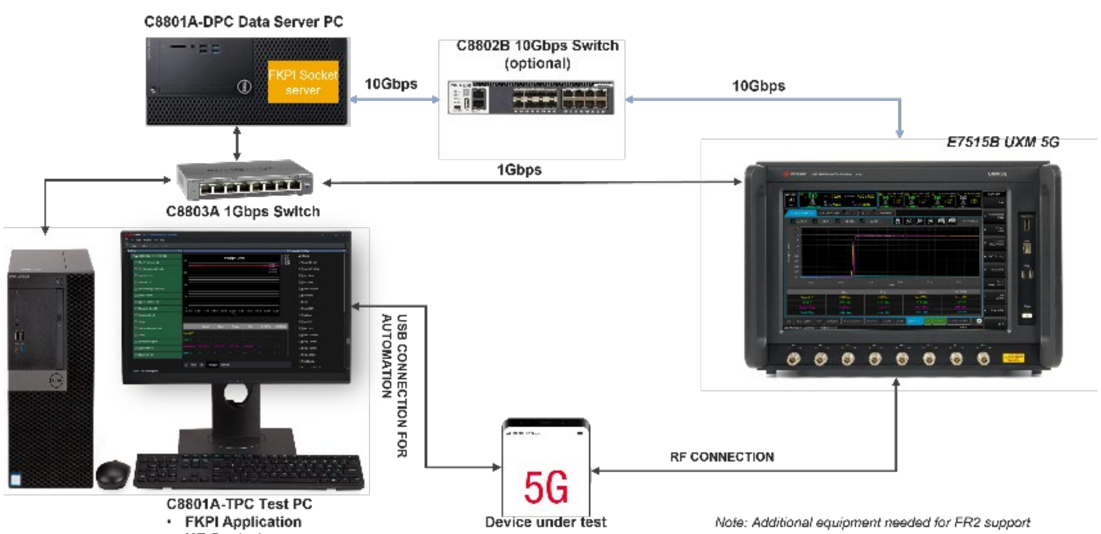

Figure 14. Typical set up for data throughput testing

The typical test setup for data throughput testing consists of the E7515B/P Base Station Emulator, which creates the signaling environment for both 4G LTE and 5GNR, along with the C8801A Data server PC from which IP test data can be streamed through the UXM and over RF to the device under test. The S8703A functional KPI software is installed on the C8801A Test PC which controls the entire test setup via LAN through the C8802B 10Gbps ethernet switch primed for the high data rates required by 5G.

Features	Specifications
Radio access technologies	5G NR (FR1 & FR2) ENDC and SA modes 4G LTE
Number of carriers	Dual Connectivity Up to 8CC Uplink & Downlink Stand Alone Up to 8CC Uplink & Downlink
MIMO scheme	DL 1x1, 2x2, 4x4 UL 1x1, 2x2, 4x4
Protocol layers tested	MAC & IP Layer Throughput
Throughput type	iPerf, FTP, UDP
Max peak throughput	10 GBPS

Battery power consumption testing

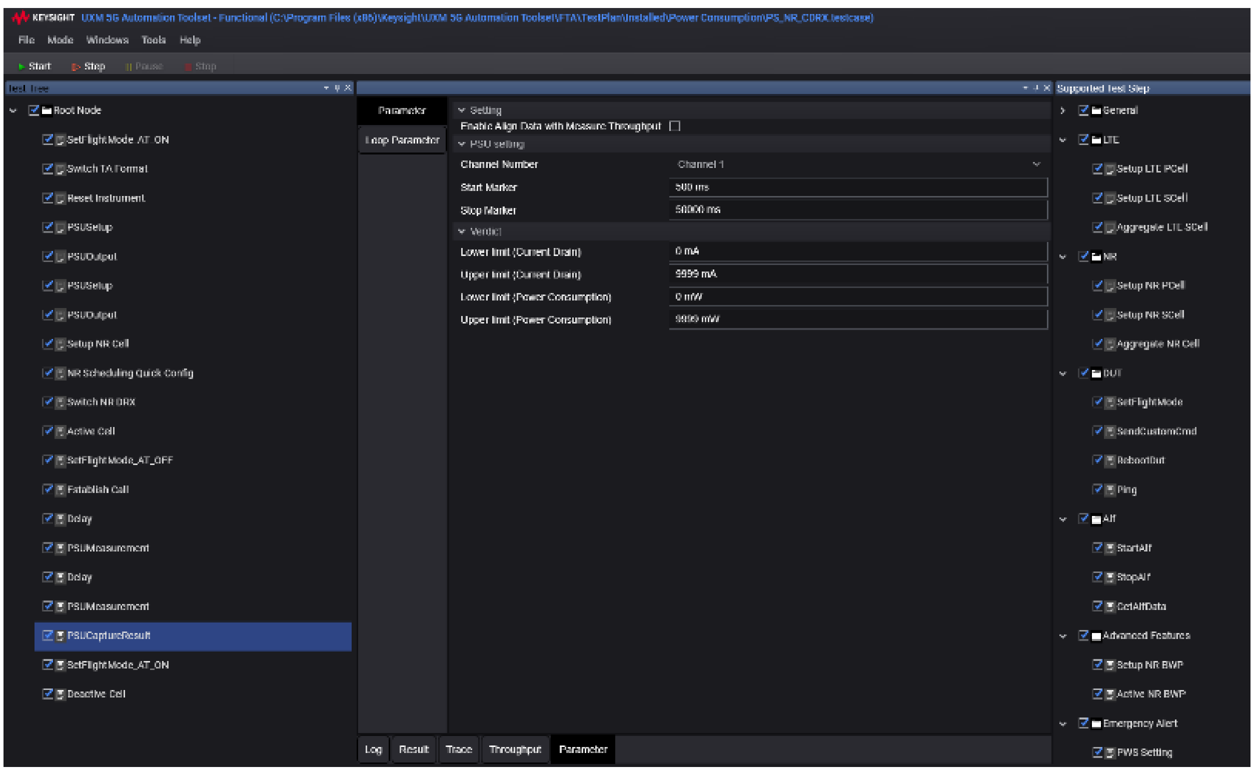

Figure 16. Battery life test configuration

The high performance and benefits of 5G come with very tasking requirements on wireless devices. The addition of millimeter wave (mmWave), higher order MIMO, higher modulation schemes, multiple carriers and wider bandwidths require devices to have an efficient system as it relates to the consumption of power from the battery. In addition to this, early devices deployed supporting dual connectivity (EN-DC) are required to monitor, transmit and receive data traffic on both LTE and 5GNR antenna thus using more power resources within the device. This makes it imperative primarily for chipset and device manufacturers to closely monitor device performance in relation to current drain while performing routine real-user tasks such as paging, standby mode, voice calls and data streaming. In addition to this, power and data hungry OTT applications are constantly running in the background of wireless devices, thus requiring thorough validation of battery performance over different diverse user profiles.

With the S8703A Functional KPI toolset engineers can easily characterize battery performance by measuring device current drain while performing usage activities in idle or connected mode. This includes paging, active voice and data calls as well as verification of the device performance while operating critical power saving features like DRX and eDRX.

Audio functional testing

As 5G networks become ubiquitous, the staged rollout of voice functionality becomes a paramount consideration for the ecosystem since not all networks deploy these features concurrently. It is imperative that current devices are tested and verified to support IMS based calls for both Voice over LTE (VoLTE) as well as Voice over New Radio (VoNR). 5G wireless networks today have different voice call deployment scenarios, all of which need to be tested and verified to ensure wireless devices can be functional in all cases.

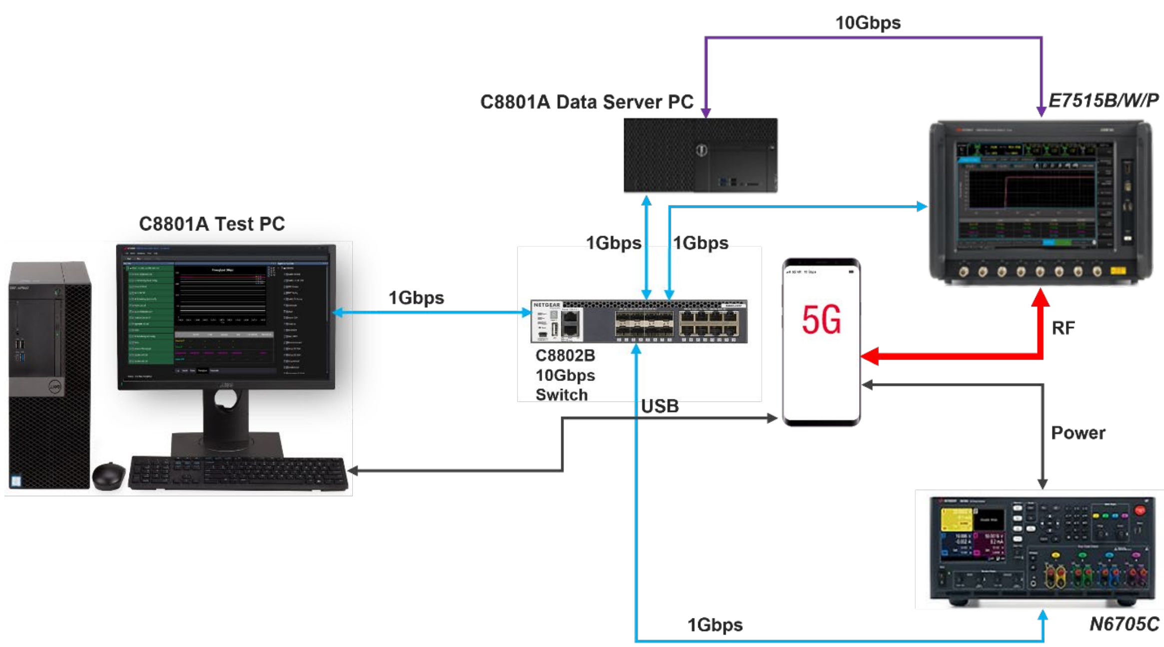

Figure 17. Typical test setup for battery life

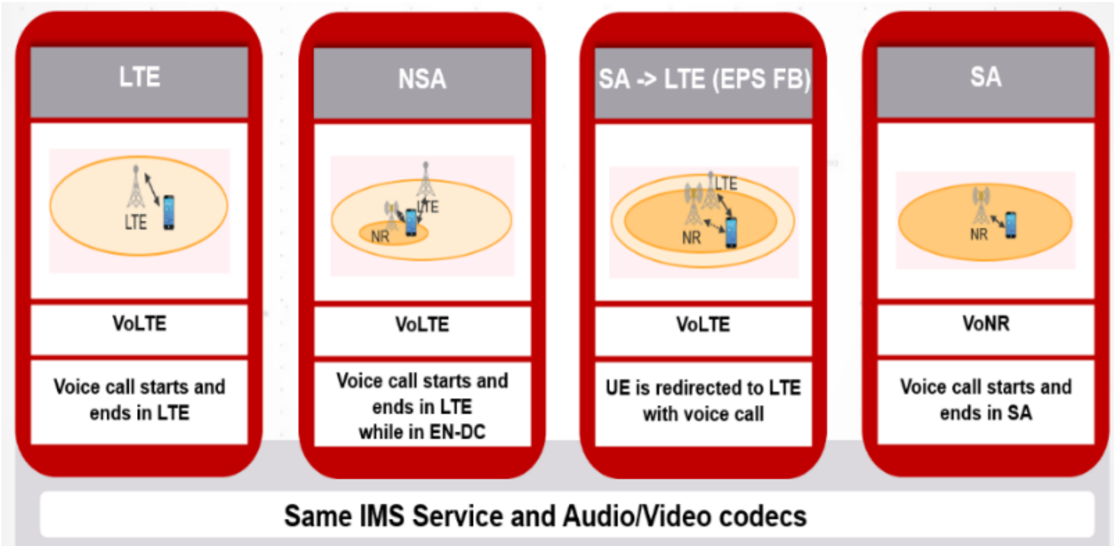

Figure 18. Audio/Voice deployment scenario for 5G NR

Current networks are either:

- Standalone 4G LTE: All voice calls starts and ends in LTE Enhanced Packet Core (EPC)
- EN-DC mode: Voice call starts and ends in LTE EPC while 5GNR device is connected to both the EPC for 4G and Next Generation Core (NGC) for 5G
- 5G Standalone mode with EPS fallback: All data traffic over 5GNR while allowing device to switch (fallback) to LTE EPC for voice calls
- 5G Standalone with VoNR: Voice calls start and end on the NGC **x**

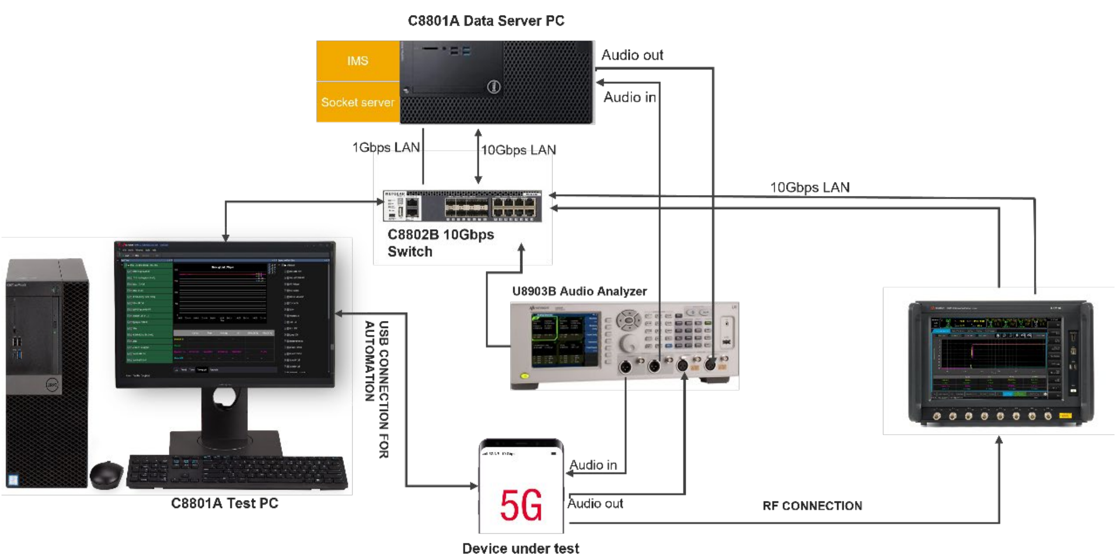

Figure 19. Test setup for VONR

These scenarios are critical to verify as they have serious performance implications in the real world. With the S8703A functional KPI toolset, users can easily setup, test and verify these use cases. With just a few mouse clicks, the user can define a test campaign that includes the key signaling parameters and conditions in an LTE or 5G network with their possible impact on the audio functionality and overall user experience. The network emulator in this case the E7515B UXM 5G can emulate an LTE and 5GNR

network according to the latest standards and specifications. The setup also provides IMS functionality to enable VoLTE and VoNR-capable devices to register, while establishing either a mobile-originated or

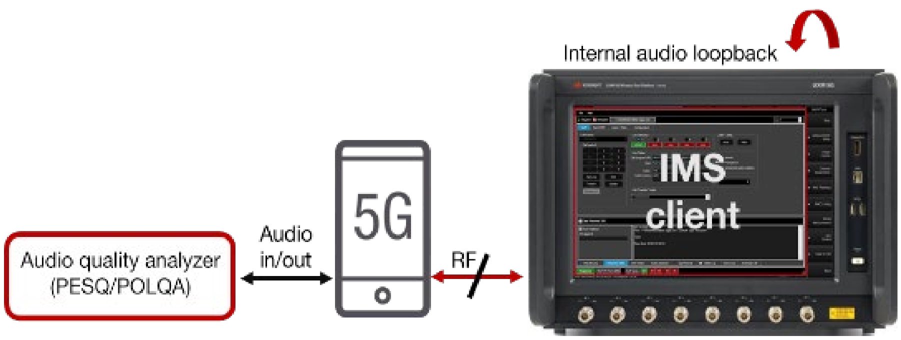

Figure 20. Typical test for Audio test

mobile-terminated voice call based on IMS. . It's also critical to be able to verify functionality with the different audio codecs such as the Adaptive Multi-Rate (AMR) codec for wideband and narrowband (AMR-WB, AMR-NB) and their respective codec rates.

Mobility testing

For 5G networks, reliability is one of the key tenets for successful deployment and adoption. Devices must be able to seamlessly move between different cell sectors, different bands, bandwidths, networks and still maintain connectivity while either in idle mode or active data connection. As operators continue to build out their 5G networks, multi RAT deployment strategies are a must, so devices must be able to support standalone and dual connectivity handovers and redirections for seamless mobility between elements of the 5G network as well as Inter-RAT scenarios.

The S8703A Functional KPI toolset along with the E7515B UXM5G:

- Offers a compact test setup to verify mobility scenarios in a single box.

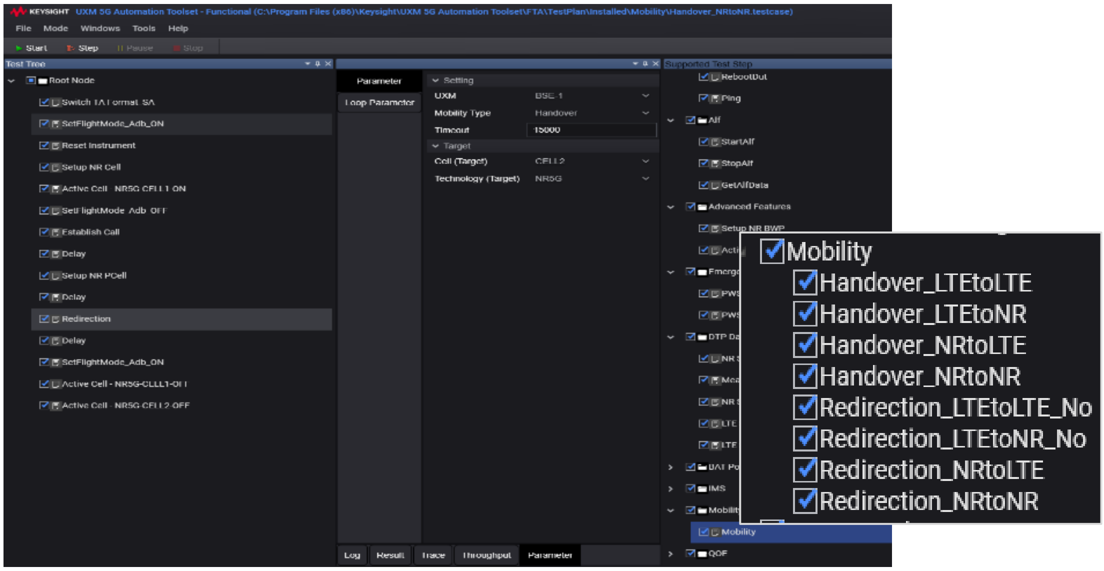

Figure 21. Supported mobility scenarios

- Leverages the multi-RAT functionality of the UXM to quickly setup standalone or Dual connectivity test and verify the seamless handover scenarios while in an active data call.
- Perform manual or event based (future) handovers
- Set up to 8 target cell for handover or redirections

Figure 22. Mobility test

Field-to-Lab testing

The S8703A Functional KPI toolset allows the users to extract the scheduling, configuration and parameters from field log collected by the UE and apply it for functional performance tests on the UXM 5G network emulator effectively.

Functional KPI toolset has a custom file parser that enables the users to import the field log.

Users can

- Load cell configuration.
- Apply scheduling to different tests
- Configure target cell with RRC Reconfiguration.
- Change PHY later parameters

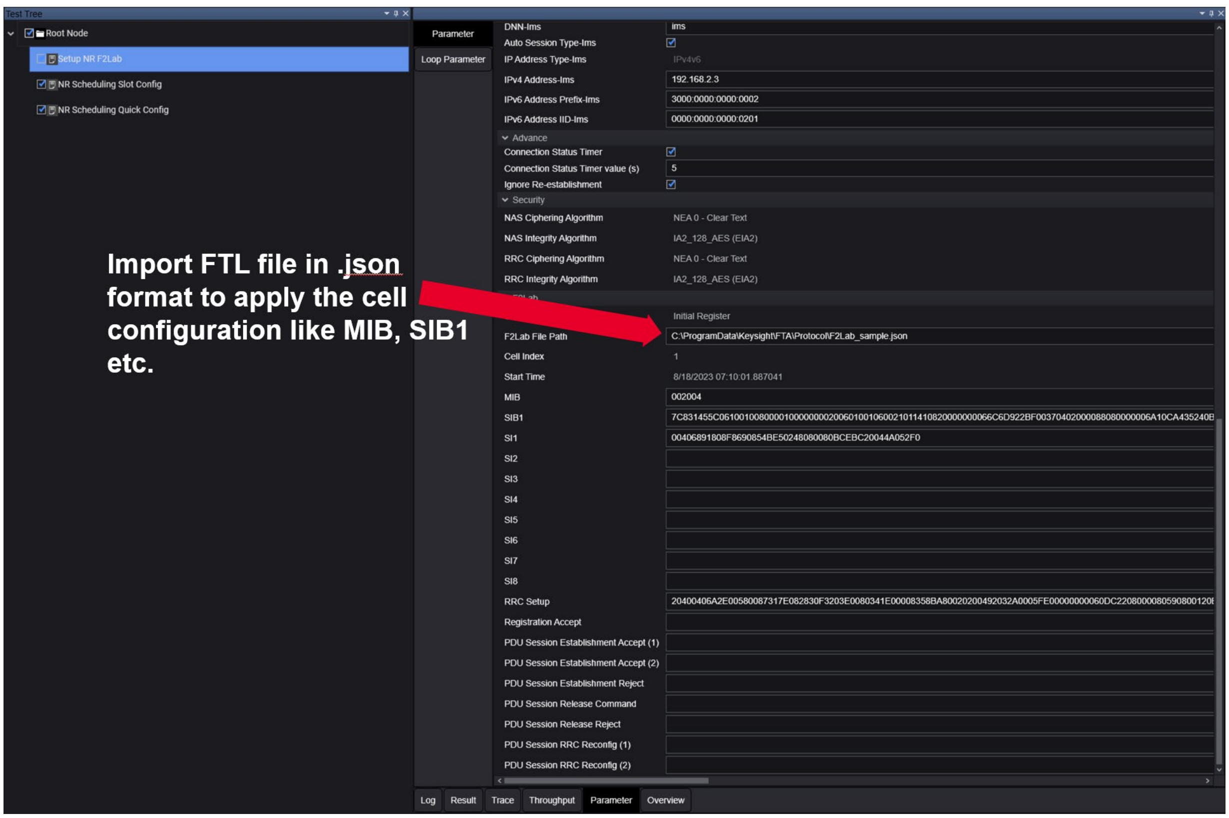

Figure 23. Field-to-lab testing using Functional KPI

Emergency services testing

Verify CMAS and ETWS

The S8703 Functional KPI toolset allows users to quickly verify critical 3GPP based public warning system right from the intuitive graphical user interface in just a few clicks.

- Test device capability to receive and decode public warning broadcast messages
- Support for Commercial Mobile Alert System (CMAS) & Earthquake and Tsunami Warning System (ETWS)
- Easily customize LTE & NR messages in different languages and coding schemes.
- Configure geographical scope of messages; *Cell wide (immediate), PLMN wide, Tracking area wide and Cell side.*

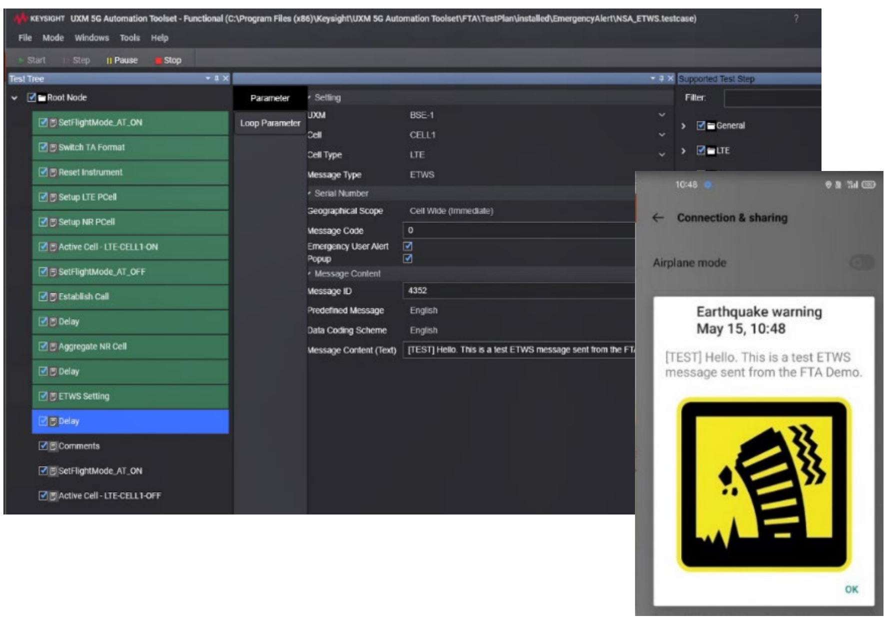

Figure 24. ETWS testing using Functional KPI

Dual UE and Dual SIM testing

Most new mobile devices support the provisioning of both physical and embedded SIM in a single device thus increasing the need to ensure proper coexistence and functionality when both systems are active. Common issues around 5G NR aggregation, registration and throughput can easily be detected using the S8703A functional KPI toolset along with the UXM 5G.

- Support for Dual UE testing, enabling testing of two devices simultaneously.
- Support for Dual SIM Dual Standby (DSDS) and Dual SIM Dual Active (DSDA) modes.
- Create bespoke scripts for user defined test scenarios.
- Get up and running with sample test scenarios.

Verify voice, data, and messaging in both idle and connected state

Figure 25. Dual SIM testing with Functional KPI

Dual SIM Dual Standby (DSDS)

Model	SIM1	SIM2
Data Performance	DL Peak Rate	Idle
	UL Peak Rate	Idle
	Bidirectional Peak Rate	Idle
	Idle	DL Peak Rate
	Idle	UL Peak Rate
	Idle	Bidirectional Peak Rate
Voice Function	VoLTE	Idle
	Idle	VoLTE

Dual SIM Dual Active (DSDA)

Model	SIM1	SIM2
Data Performance	Voice Call	DL Peak Rate
	Voice Call	UL Peak Rate
	Voice Call	Bidirectional Peak Rate
	DL Peak Rate	Voice Call
	UL Peak Rate	Voice Call
	Bidirectional Peak Rate	Voice Call
Voice Function	Voice Call	Mobility
	Mobility	Voice Call
	Voice Call	SMS
	SMS	Voice Call
	Voice Call	Incoming call
	Incoming call	Voice Call

Figure 26. Dual SIM test scenarios

With the latest enhancements made to UXM 5G platform and the network emulation solutions, the users can now test dual UE or dual SIM testing with **only one UXM 5G network emulator.**

Previously, it required the user of the full capacity of two UXM 5G units and required each of them to be setup individually, leading to a longer setup time and possible errors. Users can measure power consumption, audio quality, and throughput for each device/SIM.

This enhancement delivers increased test capacity and optimized test efficiency to the users’ testing workflow.

Figure 27. Dual SIM testing with FKPI

Figure 28. Dual UE testing with FKPI

Keysight's Network Emulation Solutions (NES) portfolio offers a range of RF test solutions, covering the entire device development workflow.

Keysight's 5G RF Solutions

Keysight's Network Emulation Solutions (NES) portfolio offers a range of RF test solutions, covering the entire device development workflow.

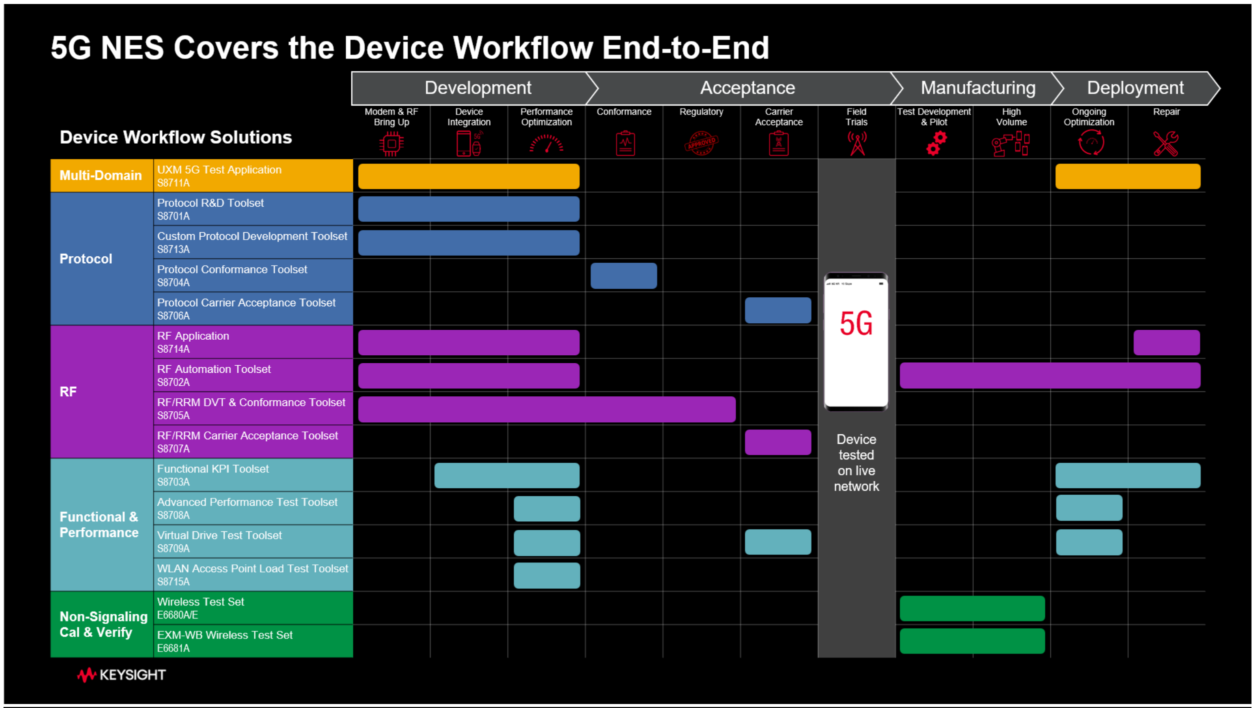

Figure 29. Network emulation solutions

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Keysight enables innovators to push the boundaries of engineering by quickly solving design, emulation, and test challenges to create the best product experiences. Start your innovation journey at www.keysight.com.